# Evaluation runs: raft (raft-rs)

generated 2026-11-05 from brain-data bb4df53ad 2026-11-05 15:31:21 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{spec}$, $G_{impl}$ = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G^{expl}_{spec}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair correct, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G^{expl}_{spec}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 24 | 1323550 | conforms | 24.3 | 29 | 1 h, 250 GB | 4 | 27 | brain01, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, reorder 1 1 2 fifo, reorder 1 2 1 fifo, correct 1 2 1 fifo, reorder 2 1 1 fifo, correct 2 1 1 fifo, correct 2 2 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo.

## stateful

### stateful: pair correct, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 1,1,1 | 24 | 1323550 | conforms | 23.9 | 29 | 1 h, 250 GB | brain01, 2026-11-02 |
| 1,2,1 | 24 | 12486268 | conforms | 258.6 | 62 | 48 h, 250 GB | brain01, 2026-11-05 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, correct 2 1 1 fifo, reorder 2 1 1 fifo, correct 2 2 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo, reorder 1 1 2 fifo, reorder 1 2 1 fifo.

## offline

### offline: pair correct, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 24 | 1323550 | conforms | 22.2 | 28 | 1 h, 250 GB | 0.1 | 22.2 | 0.0 | merge | brain01, 2026-11-03 |
| 1,2,1 | 24 | 12486268 | conforms | 287.2 | 62 | 48 h, 250 GB | 0.3 | 287.2 | 0.0 | merge | brain01, 2026-11-05 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, reorder 1 1 2 fifo, reorder 2 1 1 fifo, correct 2 2 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo.

refused (an input missing at run time): reorder 1 2 1 fifo.

skipped: a smaller size had reached its limit: correct 2 1 1 fifo.

## cross offline, Java implementation against the Rust specification

### cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 24 | 483541 | conforms | 1834.1 | 35872 | 10 min, 250 GB | 0.1 | 1834.1 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Java implementation against the Rust specification).** left out by the table rules (ok): reorder 1 1 1 fifo.

planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, correct 1 2 1 fifo, correct 2 1 1 fifo.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 24 | 1323550 | conforms | 24.0 | 158 | 10 min, 250 GB | 4.1 | 22.2 | 1.8 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Rust implementation against the Java specification).** left out by the table rules (ok): reorder 1 1 1 fifo.

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 24 | 483541 | conforms | 1952.1 | 37061 | 1 h, 250 GB | | brain01, 2026-11-03 |

**Not run (cross online, Java implementation (JMC) against the Rust specification export).** left out by the table rules (ok): reorder 1 1 1 fifo.

left out by the table rules (timeout): correct 1 1 1 fifo, reorder 1 1 1 fifo.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 24 | 1323550 | conforms | 23.4 | 29 | 1 h, 250 GB | | brain01, 2026-11-02 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** left out by the table rules (ok): reorder 1 1 1 fifo.

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 483541 | 1323550 | conforms | 1834.2 | 35872 | 48 h, 250 GB | 1834.1 | 22.2 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Rust implementation against its Java twin).** refused (an input missing at run time): reorder 1 1 1 fifo.

left out by the table rules (ok): reorder 1 1 1 fifo.

planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, correct 1 2 1 fifo, correct 2 1 1 fifo.

# twins offline, Java implementation against its Rust twin

## twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 1323550 | 483541 | conforms | 1834.1 | 35872 | 48 h, 250 GB | 22.2 | 1834.1 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Java implementation against its Rust twin).** refused (an input missing at run time): reorder 1 1 1 fifo.

planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, correct 1 2 1 fifo, correct 2 1 1 fifo.

# twins online, Rust implementation (TraceForge) against the Java twin's export

## twins online, Rust implementation (TraceForge) against the Java twin's export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 1,1,1 | 483541 | 1323550 | conforms | 44.5 | 28 | 48 h, 250 GB | | brain01, 2026-11-03 |

**Not run (twins online, Rust implementation (TraceForge) against the Java twin's export).** left out by the table rules (ok): reorder 1 1 1 fifo.

planned but never scheduled (grid points beyond the ladder): correct 1 1 2 fifo, correct 1 2 1 fifo, correct 2 1 1 fifo.